Well-Ordered Sets (Ch. 9)

Definition. A linear order $\langle X, < \rangle$ is a **well-order** if every non-empty subset of X has a minimal element.

Definition. We say that a linear order $\langle X, \lhd \rangle$ has *type omega* if X is infinite and for every $x \in X$, $\text{pred}_{\langle X, \lhd \rangle}(x)$ is finite.

Fact 9.1 If $\langle X, < \rangle$ is a linear ordere of type omega, then $\langle X, < \rangle$ is a well-order.

Lemma 9.2. If $\langle X, < \rangle$ is a well-order and $A, B \subseteq X$ are downwards closed with $\langle A, < \rangle \cong \langle B, < \rangle$, then $A = B$.

Corollary 9.3. If $\langle X, < \rangle$ is a well-order with $x < x'$, then $\langle \text{pred}_{\langle X, < \rangle}(x'), < \rangle \not\cong \langle \text{pred}_{\langle X, < \rangle}(x), < \rangle$.

Corollary 9.4. If $\langle X, < \rangle$ is a well-order, then for any $x \in X$, $\langle \text{pred}_{\langle X, < \rangle}(x), < \rangle \not\cong \langle X, < \rangle$.

Lemma 9.5. If $\langle X, < \rangle$ and $\langle Y, \prec \rangle$ are isomorphic well-orders, then the isomorphism between them is unique.

Theorem 9.6 (Fundamental Theorem of Well-Orders). If $\langle X, < \rangle$ and $\langle Y, \prec \rangle$ are well-orders, then exactly one of the following holds:

1. $\langle X, < \rangle \cong \langle Y, \prec \rangle$
2. $\exists x \in X$ such that $\langle \text{pred}_{\langle X, < \rangle}(x), < \rangle \cong \langle Y, \prec \rangle$
3. $\exists y \in Y$ such that $\langle X, < \rangle \cong \langle \text{pred}_{\langle Y, \prec \rangle}(y), \prec \rangle$

Ordinals (Ch. 10)

Definition 10.1 A set α is an **ordinal** if it is transitive and well-ordered by $\in$.

Fact 10.3 $\mathbb{N}$ is an ordinal. Every $n \in \mathbb{N}$ is also an ordinal.

Theorem 10.4. For ordinal x:

1. $\forall y \in x[y$ is an ordinal $\wedge\, y = \text{pred}_{\langle x, \in \rangle}(y)]$
2. If y is an ordinal and $\langle x, \in \rangle \cong \langle y, \in \rangle$, then $x = y$
3. If y is an ordinal, exactly one holds: $x \in y$, $x = y$, or $y \in x$
4. If y, z are ordinals, $x \in y$, and $y \in z$, then $x \in z$
5. If $C \neq \emptyset$ is a class of ordinals, then $\exists y \in C \forall z \in C[y \in z \vee y = z]$

Definition 10.5 ORD dneotes the class of all ordinals.

Theorem 10.6 (Burali-Forti). ORD is not a set.

Lemma 10.7. Every transitive set of ordinals is an ordinal.

Theorem 10.8. For any well-ordered set $\langle X, < \rangle$, there exists a unique ordinal α such that $\langle X, < \rangle \cong \langle \alpha, \in_\alpha \rangle$.

Definition 10.11 For any well-ordered set $\langle X, < \rangle$, $\text{otp}(\langle X, < \rangle)$ is the unique ordinal α such that $\langle X, < \rangle \cong \langle \alpha, \in_\alpha \rangle$.

Lemma 10.13. For ordinals α, β: $\alpha \leq \beta \Leftrightarrow \alpha \subseteq \beta$.

Lemma 10.14. For a non-empty set of ordinals A, $\min(A) = \bigcap A$. For any set of ordinals A, $\sup_{\text{ORD}}(A) = \bigcup A$.

Lemma 10.15. For any α, $S(\alpha)$ is an ordinal, $\alpha < S(\alpha)$, and $\forall \beta[\beta < S(\alpha) \Leftrightarrow \beta \leq \alpha]$.

Definition 10.16. α is a **successor ordinal** if $\exists \beta[\alpha = S(\beta)]$. α is a **limit ordinal** if $\alpha \neq 0$ and α is not a successor.

Lemma 10.17. An ordinal α is a natural number iff $\forall \beta \leq \alpha[\beta = 0 \vee \beta$ is a successor ordinal$]$.

Convention 10.18 ω denotes the set of natural numbers: $\omega = \mathbb{N}$.

Theorem 10.19 (Transfinite Induction). If $\forall \alpha \in \text{ORD}[\forall \beta < \alpha[P(\beta)] \Rightarrow P(\alpha)]$, then $\forall \alpha \in \text{ORD}[P(\alpha)]$.

Definition 10.20 Let **FOD** denote the class of all functions whose domain is some ordinal. $\mathbf{FOD} = \{\sigma : \sigma$ is a function $\wedge\, \exists \alpha \in \mathbf{ORD}[\text{dom}(\sigma) = \alpha]\}$.

Theorem 10.21 (Transfinite Recursion). For any extender $\mathbf{E} : \mathbf{FOD} \to \mathbf{V}$, there exists a unique function $\mathbf{F} : \mathbf{ORD} \to \mathbf{V}$ satisfying $\forall \alpha \in \mathbf{ORD}[\mathbf{F}(\alpha) = \mathbf{E}(\mathbf{F} \restriction \alpha)]$.

Ordinal Arithmetic (Ch. 11)

Defintion 11.1. Let $\langle X, <_X \rangle$ and $\langle Y, <_Y \rangle$ be well-orders. Define $X \oplus Y$ to be the set $(\{0\} \times X) \cup (\{1\} \times Y)$. Define $<_{X \oplus Y}$ on $X \oplus Y$ by the following clauses:

1. $\forall x, x' \in X\, [\langle 0, x \rangle <_{X \oplus Y} \langle 0, x' \rangle \Leftrightarrow x <_X x']$;
2. $\forall y, y' \in Y\, [\langle 1, y \rangle <_{X \oplus Y} \langle 1, y' \rangle \Leftrightarrow y <_Y y']$;
3. $\forall x \in X\, \forall y \in Y\, [\langle 0, x \rangle <_{X \oplus Y} \langle 1, y \rangle]$.

Definition 11.2. (Ordinal Addition). For ordinals α, β, $\alpha + \beta = \text{otp}(\langle \alpha \oplus \beta, <_{\alpha \oplus \beta} \rangle)$.

Lemma 11.4 Let $\langle X, <_X \rangle$, $\langle Y, <_Y \rangle$, $\langle Z, <_Z \rangle$ be well-orders. Suppose that $A, B \subseteq Z$. Assume that $A \cup B = Z$ and that $\forall a \in A\, \forall b \in B\, [a <_Z b]$. Then if $\langle A, <_Z \rangle$ is isomorphic to $\langle X, <_X \rangle$ and $\langle B, <_Z \rangle$ is isomorphic to $\langle Y, <_Y \rangle$, then $\langle Z, <_Z \rangle$ is isomorphic to $\langle X \oplus Y, <_{X \oplus Y} \rangle$.

Lemma 11.5. For ordinals α, β, γ:

1. $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$
2. $\alpha + 0 = \alpha$
3. $\alpha + 1 = S(\alpha)$
4. $\alpha + S(\beta) = S(\alpha + \beta)$
5. If β is a limit ordinal, $\alpha + \beta = \sup\{\alpha + \xi : \xi < \beta\}$

Definition (Ordinal Multiplication). For ordinals α, β, $\alpha \cdot \beta = \text{otp}(\langle \beta \times \alpha, <_{\alpha \cdot \beta} \rangle)$ where $<_{\alpha \cdot \beta}$ is the dictionary order.

Lemma 11.8. Suppose α, β, γ are ordinals. Suppose $A \subseteq \gamma$ and that $\langle A, \in \rangle$ is isomorphic to $\langle \beta, \in \rangle$. Then $\langle A \times \alpha, <_{\alpha \cdot \gamma} \rangle$ is isomorphic to $\langle \beta \times \alpha, <_{\alpha \cdot \beta} \rangle$.

Lemma 11.9. For ordinals α, β, γ:

1. $\alpha \cdot (\beta \cdot \gamma) = (\alpha \cdot \beta) \cdot \gamma$
2. $\alpha \cdot 0 = 0$
3. $\alpha \cdot 1 = \alpha$
4. $\alpha \cdot S(\beta) = \alpha \cdot \beta + \alpha$
5. If β is a limit ordinal, $\alpha \cdot \beta = \sup\{\alpha \cdot \xi : \xi < \beta\}$
6. $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$

Definition (Ordinal Exponentiation). For fixed α, define α^β by recursion:

1. If $\alpha = 0$, then $\alpha^0 = 0$; if $\alpha > 0$, then $\alpha^0 = 1$
2. $\alpha^{\beta+1} = \alpha^\beta \cdot \alpha$
3. If β is a limit ordinal, then $\alpha^\beta = \sup\{\alpha^\xi : \xi < \beta\}$

Properties:

1. $\alpha^{\beta+\gamma} = \alpha^\beta \cdot \alpha^\gamma$ for $\alpha > 0$
2. For any $\alpha > 0$, $\alpha \cdot \omega > \alpha$
3. $\alpha < \beta \Rightarrow \gamma + \alpha < \gamma + \beta$ and $\alpha + \gamma \leq \beta + \gamma$
4. If $\alpha \geq \omega$ is an ordinal, then $1 + \alpha = \alpha$
5. If $\gamma > 0$, then $\alpha < \beta \Rightarrow \gamma \cdot \alpha < \gamma \cdot \beta$ and $\alpha \cdot \gamma \leq \beta \cdot \gamma$

Cardinals and Cardinal Arithmetic (Ch. 12)

Definition. A set X is **well-orderable** if there exists a relation $< \subseteq X \times X$ such that $\langle X, < \rangle$ is a well-order.

Definition. For a well-orderable set X, the **cardinality** of X, written $|X|$, is the minimal ordinal α such that $\alpha \approx X$.

Definition. α is a **cardinal** if $|\alpha| = \alpha$.

Lemma 12.5. If $|\alpha| \leq \beta \leq \alpha$, then $|\beta| = |\alpha|$.

Lemma 12.6. A set is finite iff $|X| < \omega$. A set is countable iff $|X| \leq \omega$.

Definition. For cardinals κ and λ:

1. $\kappa \oplus \lambda = |(\{0\} \times \kappa) \cup (\{1\} \times \lambda)|$
2. $\kappa \otimes \lambda = |\kappa \times \lambda|$

Lemma 12.8. Every infinite cardinal is a limit ordinal.

Theorem 12.9 (Fundamental Theorem of Cardinal Arithmetic). If κ is an infinite cardinal, then $\kappa \otimes \kappa = \kappa$.

Corollary 12.10. For infinite cardinals κ, λ: $\kappa \oplus \lambda =$

$\kappa \otimes \lambda = \max\{\kappa, \lambda\}$.

Theorem 12.11. For every set X there is a cardinal α such that there is no 1-1 function $f : \alpha \to X$.

Corollary 12.13. For every $\beta \in \text{ORD}$, there exists a cardinal $\alpha > \beta$.

Lemma 12.15. Let A be a set of cardinals. Then $\bigcup A$ is a cardinal.

Definition. For each $\alpha \in \text{ORD}$, α^+ is the least cardinal strictly greater than α.

Lemma 12.17. Suppose $\mathbf{F} : \mathbf{ORD} \to \mathbf{ORD}$ is a function such that $\forall \alpha, \beta \in \mathbf{ORD}[\alpha < \beta \Rightarrow \mathbf{F}(\alpha) < \mathbf{F}(\beta)]$. Then $\forall \beta \in \mathbf{ORD}[\beta \leq \mathbf{F}(\beta)]$.

Definition. Define $\langle \omega_\alpha : \alpha \in \text{ORD} \rangle$ by:

1. $\omega_0 = \omega$
2. $\omega_{S(\alpha)} = \omega_\alpha^+$
3. If α is a limit ordinal, then $\omega_\alpha = \sup\{\omega_\xi : \xi < \alpha\}$

Definition. $\aleph_\alpha$ denotes ω_α.

Lemma 12.20.

1. $\alpha < \beta \Rightarrow \aleph_\alpha < \aleph_\beta$
2. Every infinite cardinal equals $\aleph_\alpha$ for some $\alpha \in \text{ORD}$

Definition. Let X be any set. We say that F is a choice function on X if F is a function, $\text{dom}(F) = X \setminus \{\emptyset\}$, and $\forall a \in X \setminus \{\emptyset\} [F(a) \in a]$.

Theorem 12.22 (Zomelo) The following are equivalent for a set X:

1. X is well-orderable;
2. there exists a choice function on $\mathcal{P}(X)$.

Theorem 12.26 (Equivalent Forms of AC). The following are equivalent:

1. Cartesian product of non-empty sets is non-empty
2. Every set has a choice function
3. Every set is well-orderable
4. For any two sets X, Y, either $X \preceq Y$ or $Y \preceq X$
5. For every set X, there is an ordinal α and a 1-1 function $f : X \to \alpha$
6. For every set X, there is a cardinal κ such that $X \approx \kappa$

Definition (Cardinal Exponentiation). $\kappa^\lambda = |\{f : f \text{ is a function} \wedge \text{dom}(f) = \lambda \wedge \text{ran}(f) \subseteq \kappa\}|$.

Lemma 12.30. For cardinals κ, λ, θ:

1. $\left(\kappa^\lambda\right)^\theta = \kappa^{(\lambda \otimes \theta)}$
2. $\kappa^\lambda \otimes \kappa^\theta = \kappa^{(\lambda \oplus \theta)}$

Definition. Define $\langle \beth_\alpha : \alpha \in \text{ORD} \rangle$ by:

1. $\beth_0 = \omega$
2. $\beth_{S(\alpha)} = 2^{\beth_\alpha}$
3. If α is a limit ordinal, then $\beth_\alpha = \sup\{\beth_\xi : \xi < \alpha\}$

Definition. The Generalized Continuum Hypothesis (GCH) states that $\forall \alpha \in \text{ORD}[\beth_\alpha = \aleph_\alpha]$. The Continuum Hypothesis (CH) states that $\beth_1 = \aleph_1$.

From Chapter 13

Theorem 13.3. The following are equivalent:

1. AC
2. For any set A and $F \subseteq \mathcal{P}(A)$, if F has finite character, then for every $X \in F$, there exists $Y \in F$ such that $X \subseteq Y$ and Y is maximal in $\langle F, \subset \rangle$ (Teichmüller-Tukey lemma)
3. Every chain in every partial order is contained in a maximal chain (Hausdorff's maximal chain theorem)
4. If $\langle X, < \rangle$ is any partial order where every chain has an upper bound, then $\langle X, < \rangle$ has a maximal element (Zorn's lemma)